ModelIt!

Teacher's Guide

Carbon Capture at Scale: Can We Vacuum the Atmosphere?

Modeling the Engineering and Economics of Removing CO2 from the Sky

Quick Reference

Grade Level	Grade 10L3
NGSS Standard	HS-ESS2-6, HS-PS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS2-6, HS-PS1-4

Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere. Develop a model to illustrate that the release or absorption of energy from a chemical reaction system depends upon the changes in total bond energy.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use quantitative models to calculate net CO_2 removal, energy balances, and economic viability of carbon capture systems at various scales.
Disciplinary Core Idea	**ESS2.D Weather and Climate / PS1.B Chemical Reactions** — Students model the carbon cycle at global scale and analyze the chemical reactions involved in CO_2 capture and sorbent regeneration, including energy changes.
Crosscutting Concept	**Scale, Proportion, and Quantity** — Students grapple with the enormous scale mismatch between current capture capacity (thousands of tons) and climate-relevant removal (billions of tons), analyzing what scaling requires.

CAST Testing Alignment

MULTIPLE CHOICE:

A direct air capture facility powered by natural gas removes 10,000 tons of CO_2 per year but emits 4,000 tons from energy generation. What is the facility's net removal rate, and what does your model predict would happen if the facility switched to solar power?

CONSTRUCTED RESPONSE:

Using your model, argue whether direct air capture should be considered a primary climate solution or a supplementary technology. Address scale, cost, energy requirements, and the relationship between capture and emission reduction in your response.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Humanity has added over 1.5 trillion tons of CO_2 to the atmosphere since the Industrial Revolution, raising concentrations from 280 to 420+ parts per million. Even if all emissions stopped today, the CO_2 already in the atmosphere will continue warming the planet for centuries. Carbon capture technology offers the possibility of actively removing this legacy CO_2 — but the gap between what is technically possible and what is needed to impact climate is staggering.

How Direct Air Capture Works

Direct air capture (DAC) uses large fans to draw ambient air across chemical sorbents that selectively bind CO_2 molecules. Solid sorbent systems (like Climeworks' technology) use amine-functionalized filters that capture CO_2 at ambient temperature and release it when heated to 80-120 degrees Celsius. Liquid solvent systems (like Carbon Engineering's approach) bubble air through potassium hydroxide solution, then use a calcination process at 900 degrees Celsius to release pure CO_2. Both approaches require significant energy for sorbent regeneration and CO_2 compression.

The Scale Challenge

Global CO_2 emissions exceed 36 billion tons per year. The world's largest DAC facility (Climeworks' Orca in Iceland) captures 4,000 tons per year. To capture just 1% of annual emissions would require 90,000 Orca-sized facilities. To make a meaningful climate impact by removing 10 billion tons per year by 2050 would require building an industry roughly the size of the current global oil and gas sector — but in reverse. This scale challenge is the most sobering aspect of carbon capture.

Energy and Economics

Current DAC costs range from $400-1,000 per ton of CO_2 captured, depending on technology and energy source. The energy required is 6-10 GJ per ton — meaning capturing 1 billion tons would consume roughly 1% of global energy production. If that energy comes from fossil fuels, the capture process emits substantial CO_2, potentially negating 30-50% of what it captures. Only when powered by clean energy does DAC achieve its full removal potential. Cost projections suggest $100-200 per ton may be achievable by 2050 with technology improvements and scale.

Geological Storage and Permanence

Captured CO_2 must be permanently stored to have climate benefit. Geological sequestration injects compressed CO_2 into deep formations: saline aquifers (porous rock saturated with saltwater), depleted oil and gas reservoirs, or basalt formations where CO_2 mineralizes into carbonate rock within years. Global geological storage capacity is estimated at 2,000+ gigatons — far more than needed. The key challenges are ensuring long-term integrity (no leakage over centuries), monitoring stored CO_2, and building injection infrastructure near capture sites.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of a carbon capture system: CO2 concentration, capture rate, sorbent efficiency, energy input, storage capacity, reaction temperature, regeneration cost, transport distance, geological sequestration, and carbon credit value.

E **Establish Relationships**

Students map relationships showing that energy input drives both capture rate and regeneration cost, that transport distance and geological sequestration determine storage feasibility, and that carbon credit value must exceed total costs for economic viability.

V **Visualize & Model**

Students build a computational model showing the ten-component carbon capture system with energy-emissions coupling, economic viability thresholds, and scale constraints.

E **Evaluate Outcomes**

Students run current technology, clean energy powered, and climate-scale scenarios to identify the conditions under which carbon capture transitions from expensive demonstration to viable climate solution.

R **Revise & Extend**

Students add sorbent lifetime, water consumption, or CO2 utilization revenue to model more complete techno-economic dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"We have put 1.5 trillion tons of CO2 into the atmosphere. What if we could vacuum it back out? That is not science fiction — this facility is doing it right now. But there is a catch."*

Watch For

Show image of Climeworks' Orca facility in Iceland. Reveal the catch: it removes 4,000 tons per year; we emit 36 billion. Let students process the scale gap.

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling whether technology can actually undo what we have done to the atmosphere — and at what cost."*

Watch For

Have students read objectives. Pre-teach 'sorbent' and 'geological sequestration.' Use a baking soda-vinegar analogy for chemical CO2 capture.

Slide 3: Big Question

3 min

Say: *"Some say carbon capture is essential. Others say it is a fossil fuel industry excuse to keep polluting. Your model will help you decide who is right."*

Watch For

Quick-write: Students position themselves — is carbon capture the solution, a distraction, or something in between? Take initial positions to revisit later.

Slide 4: LEVER Framework

2 min

Say: *"We are modeling a system with ten variables spanning chemistry, engineering, geology, and economics. The answer depends on ALL of them working together."*

Watch For

Preview LEVER steps. Emphasize the interdisciplinary nature — this is where science meets economics meets policy.

Slide 5: Activity 1: Components

8 min

Say: *"Ten variables. Three you control through engineering choices. Seven the physics and economics determine. The question is whether the math works."*

Watch For

Guide students through all ten components. Emphasize the energy-emissions paradox upfront — this is the model's central tension.

Slide 6: Activity 2: Connections

8 min

Say: *"Here is the paradox: you need energy to capture CO2, but if that energy produces CO2, you might be making the problem worse. Trace that loop."*

Watch For

Help students map the energy-emissions coupling. Highlight the economic viability threshold where carbon credit value must exceed total cost.

Slide 7: Activity 3: Simulation

10 min

Say: *"Does carbon capture actually help the climate? It depends entirely on HOW you do it. Let us find out."*

Watch For

Students predict net removal for each scenario. Run simulations. The clean energy scenario versus fossil energy scenario should be eye-opening.

Slide 8: Discoveries

5 min

Say: *"Carbon capture is real, it works, and it is necessary. But it cannot replace emission reduction — the scale gap is too enormous. Both strategies are required."*

Watch For

Lead discussion revisiting initial positions. How did the model change students' views? Discuss the role of carbon capture as one tool in a larger climate strategy.

Slide 9: STEM Challenge

12 min

Say: *"A venture fund has 100 million dollars. Design a facility that actually makes money removing CO2 from the atmosphere. Can you make the economics work?"*

Watch For

Groups design commercially viable capture facilities. Present with economic projections from model data. Class evaluates which facility design best balances cost, scale, and climate impact.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** CO_2 Concentration, Energy Input, Transport Distance
- **INTERNAL:** Capture Rate, Sorbent Efficiency, Storage Capacity, Reaction Temperature, Regeneration Cost, Geological Sequestration, Carbon Credit Value

Relationships

- **Energy Input to Capture Rate:** POSITIVE (+) — More energy powers more fans and sorbent regeneration cycles, increasing the volume of air processed and the mass of CO_2 captured per day. Energy is the primary driver of capture throughput.
- **Sorbent Efficiency to Regeneration Cost:** NEGATIVE (-) — Higher sorbent efficiency means more CO_2 is captured per regeneration cycle, reducing the number of cycles needed and the thermal energy consumed. More efficient sorbents lower the per-ton cost of regeneration.
- **Carbon Credit Value to Capture Rate:** POSITIVE (+) — Higher carbon credit prices create economic incentive to invest in larger facilities and more efficient technology, increasing capture capacity. When credits are too cheap, facilities cannot cover operating costs and scale stagnates.

Key Discoveries

- The energy source powering a carbon capture facility determines whether it is a climate solution or a climate problem — fossil-powered capture can emit more CO_2 than it removes
- At current costs of $400-1,000 per ton, direct air capture is far more expensive than most emission reduction strategies — planting trees captures CO_2 at $5-50 per ton, though with less permanence
- Scaling to climate-relevant levels (gigatons per year) requires building an industry comparable in size to the current oil and gas sector — in reverse
- Carbon credits create a necessary economic driver, but the market price must reach $200+ per ton sustainably for direct air capture to be economically viable without subsidies

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Carbon capture means we do not need to reduce emissions"	Carbon capture and emission reduction are complementary, not substitutional. The scale mismatch is enormous: we emit 36+ billion tons per year and capture thousands. Even optimistic projections suggest capture reaching 10 billion tons per year by 2050 — still less than a third of current emissions. Carbon capture is essential for removing legacy CO2 and addressing sectors where emissions are hardest to eliminate (aviation, cement, steel), but it cannot compensate for continued large-scale fossil fuel burning. Framing it as an alternative to emission reduction is both scientifically and economically unrealistic.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Planting trees is always better than technology-based capture"	Trees are excellent carbon sinks at $5-50 per ton, but they have limitations: they require large land areas (competing with agriculture), take decades to reach full sequestration capacity, are vulnerable to wildfires and disease that release stored carbon, and saturate as forests mature. Direct air capture is more expensive but offers permanent geological storage, requires less land, and captures CO2 regardless of season or weather. The optimal strategy uses both: trees for affordable, near-term removal and DAC for permanent, scalable, verifiable removal.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Captured CO2 will leak back out of underground storage"	Geological sequestration is remarkably permanent when done correctly. CO2 injected into deep saline aquifers dissolves in brine, reacts with minerals, and eventually converts to carbonate rock — a process that takes centuries but is essentially permanent. Natural CO2 reservoirs have held gas underground for millions of years. Monitoring technology can detect leaks of less than 0.01% per year. The real risk is not geological — it is regulatory: ensuring that storage sites are properly selected, operated, and monitored for the long term.